MIL-G-16959B
12 December 1967
SUPERSEDING
MIL-G-16959A
2 December 1952
(Sec 6.3)

MILITARY SPECIFICATION

GASKETS; METALLIC, BOILER, ECONOMIZER AND SUPERHEATER HANDHOLE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers metallic gaskets for use on 2-inch round handholes of boiler economizers and superheaters.

1.2 Classification. Gaskets shall be of the following sizes, as specified (see 6.2):

Size - 3/4-inch bearing surface width.
Size - 1-1/4-inches bearing surface width.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
WW-T-731 - Tubes, Steel and Open Hearth Iron, Seamless and Welded, Boiler Use.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay-Material).
PPP-B-676 - Boxes, Setup.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-T-16286 - Tubes, Steel, Seamless, Marine Boiler Application.
MIL-T-17188 - Tubes, Carbon Steel, Resistance Welded, Marine Boiler.

STANDARDS

FEDERAL
FED-STD-151 - Metals, Test Methods.

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

DRAWING

MILITARY
5000-S5101-F-841218 - Gaskets, Boiler Handhole, Steam Service.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A-178 - Electric Resistance - Welded Carbon Steel Boiler Tubes.
A-192 - Seamless Carbon Steel Boiler Tubes for High-Pressure Service.

FSC 5330

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

UNIFORM CLASSIFICATION COMMITTEE

UNIFORM FREIGHT CLASSIFICATION RULES.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Gaskets shall be manufactured from ingot iron with a maximum chemical content of designated items as shown in table I, total content of carbon, manganese, phosphorus, sulfur, and silicon not to exceed 0.14 percent; from resistance-welded tubing in accordance with class A of MIL-T-17188 or grade A of ASTM A-178; or from seamless steel tubing in accordance with class A of MIL-T-16286, composition A of WW-T-731 or ASTM A-192.

TABLE I - Chemical composition.

Composition	Carbon	Manganese	Sulfur (max.)	Phosphorus (max.)
	Percent	Percent	Percent	Percent
Ingot iron.....	0.05 (Max.)	0.10 (Max.)	0.035	0.035

3.2 Construction.

3.2.1 Gaskets of ingot iron. Gaskets manufactured from ingot iron shall be machined or ground to the required inside and outside diameter as indicated by a dial indicator. Approximately 0.015 inch of metal shall be removed on a side. The gaskets shall be reduced to proper length by removing the rounded edge on one end of the gasket by facing. Gaskets shall be machined or ground to the required dimensions and finish (see 3.5).

3.2.2 Gaskets of tubing. Gaskets manufactured from resistance-welded tubing and seamless steel tubing shall be cut to length from 2-1/16 inches outside diameter, No. 11 gage (0.120 inch min. wall) tubes. The gaskets shall be machined or ground to the required inside and outside diameters, other dimensions and finish (see 3.5).

3.3 Heat treatment (Annealing). Gaskets, regardless of material, shall be annealed by placing rough gaskets in a cast iron box and sealing box with one layer of dry asbestos cement, followed by one layer of wet asbestos cement to prevent moisture from entering the the box. The box shall then be placed in a furnace and the temperature within the box raised to from 1250° to 1350°F. The temperature within the box shall be maintained at this range of temperature for 3 hours. The heating medium shall then be shut off and the furnace doors sealed with fire clay and the gaskets allowed to cool in the box to a black heat (about 900°F.) before opening the furnace door and removing the box.

3.4 Hardness. Gaskets shall show a hardness of not more than 65 on the Rockwell B scale.

3.5 Dimensions and finish. Dimensions, tolerances and machining finish shall be as shown on Drawing 5000-S5101-F-841218 for taper plug gaskets. Burrs shall be removed from the ends of gaskets, regardless of material, and corners broken. Both ends of the gasket shall be given a 30° inside taper as shown on Drawing 5000-S5101-F-841218.

3.6 Product marking. Identification marking shall be accomplished in accordance with MIL-STD-130.

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3.7 Workmanship. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 Lot. All gaskets of the same size manufactured from the same batch of material and presented at one time shall be considered a lot for purposes of examination and tests.

4.2.2 Sampling for examination. A random sample of gaskets shall be selected from each lot of material offered for visual and dimensional characteristics with lot acceptance based on the sampling procedures of MIL-STD-105, Special Inspection Level S-1. Acceptable Quality Level (AQL) shall be 1.5 percent defective.

4.2.3 Sampling for chemical analysis. Two samples shall be selected for chemical analysis.

4.2.4 Sampling for hardness test. A random sample of gaskets shall be selected from each lot in accordance with MIL-STD-105, Special Inspection Level S-3, and shall be subjected to the Rockwell hardness specified in 4.3.3. The AQL shall be 1 percent defective.

4.3 Quality conformance inspection.

4.3.1 Visual and dimensional examination. Each of the sample gaskets selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification. Examination shall be conducted in accordance with table II. Inside diameter of all gaskets shall be checked with a "GO" and "NOT GO" plug gage. Outside diameter of all gaskets shall be checked with a "GO" and "NOT GO" snap gage, and wall thickness shall be checked with a dial inspector fixture gage for compliance with the dimensions and tolerances shown on Drawing 5000-S-5101-F-841218. Any gasket in the sample containing one or more visual or dimensional defects shall be subject to rejection, and if the number of defective gaskets in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

Table II - Classification of defects.

Categories	Defects
Critical:	None defined.
Major:	
101	Chemical content not as specified.
102	Hardness over limit of 65 on Rockwell B scale.
103	Bearing surface not as specified within tolerance of plus or minus 0.031 inch.
104	Outside diameter not 2.015 inches within tolerance of plus or minus 0.003 inch.
105	Thickness not 0.055 inch within tolerance of plus 0.003 inch, minus 0.001 inch.
106	Inside taper both ends not 30 degrees, not within 0.031 inch minimum, 0.062 inch maximum.
107	Inside and outside finish not within description of roughness height rating 125.
108	Burrs not removed from both ends of gaskets and corners not broken.
109	Preservation not as specified.

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Table II - Classification of defects. (Cont'd)

Categories	Defects
Minor:	
201	Interior packages not prepared as specified; any packaging requirement not as specified; number of maximum items per basic package not complying; insufficient room in package to prevent jamming.
202	For interior packaging, insufficient or insecure flat steel strapping or tape binding; inadequate application of components such as incomplete closure of container flaps, improper stapling or taping; broken, crushed, or otherwise damaged shipping containers; gross weight of interior packages not as specified.
203	Packing not in accordance with contract requirements; shipping containers not snug-fitting; voids not filled; filler for voids not of specified material; not furnished with fiberboard liner or pad as specified; gross weight of contents of shipping container in excess of weight specified; any nonconforming component; component missing, damaged, or otherwise defective; inadequate application of components such as incomplete closures of case liners, container flaps, loose or inadequate strappings, bulged or distorted containers.
204	Interior and exterior markings as specified, omitted, illegible, incorrect, incomplete, of improper size location, or not in accordance with contract requirements.

4.3.2 Chemical analysis. Each of the sample gaskets selected in accordance with 4.2.3 shall be analyzed by method 111 of FED-STD-151 to determine conformance with 3.1. If any sample fails to conform to 3.1, this shall be cause for rejection of the entire lot.

4.3.3 Hardness test. Each of the sample gaskets selected in accordance with 4.2.4 shall be tested by method 243 of FED-STD-151 to determine conformance with 3.4. Any sample gasket which does not meet the requirements specified in 3.4 shall be subject to rejection, and if the number of nonconforming gaskets in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample. Rejected lots shall not be resubmitted.

4.3.4 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing, and markings comply with the requirements of Section 5 of this specification. Examination shall be conducted in accordance with table II. The sample units for this examination shall be shipping containers fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects. The lot size shall be the number of shipping containers in the lot. Selection of random samples and examination shall be made at Inspection Level S-2 of MIL-STD-105. The AQL shall be 2.5 percent defective.

5. PREPARATION FOR DELIVERY

(Preparation for delivery requirements specified herein apply only to direct Government procurement. Preparation for delivery requirements between contractors and subcontractors shall be as specified in the individual order.)

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C, as specified (see 6.2).

5.1.1 Level A. Gaskets shall be preserved and packaged Method I of MIL-P-116. Only gaskets of like size shall be packaged in a unit package. Quantity per unit package shall be as specified by the procuring activity. Preservation compound shall conform to type P-2 or P-6 or MIL-P-116. Unit packages shall be packaged in intermediate containers conforming to PPP-B-566, PPP-B-665, or PPP-B-676. The weight of each intermediate packages under PPP-B-566 or PPP-B-676 shall not exceed 5 pounds. The weight of each intermediate package under PPP-B-665 shall not exceed 20 pounds. Package closures shall conform to the applicable specifications or appendix thereto.

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5.1.2 Level C. Gaskets shall be preserved and packaged by size in a manner which affords adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practices, provided the latter meets the requirements of this level.

5.2 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Gaskets packaged as specified in 5.1.1 shall be packed in fiberboard shipping containers conforming to PPP-B-636, class weather resistant. Insofar as practicable, shipping containers shall be uniform in shape and size and contain identical quantities. Gross weight shall be limited to the special requirements of the container specification. Container closure and sealing shall be in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Level B shall be the same as Level A, except the fiberboard shipping container shall conform to class domestic.

5.2.3 Level C. Gaskets packaged as specified in 5.1.2 shall be packed in a manner which affords adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This level shall comply with the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation. This level may conform to the suppliers commercial practice provided the latter meets the above requirements.

5.3 Marking. In addition to any special marking required by the contract or order, the unit packages, intermediate packages, and exterior shipping containers shall be marked in accordance with MIL-STD-129. Product marking shall be as specified in 3.6.

6. NOTES

6.1 Intended use.

6.1.1 Size 3/4-inch gaskets (Federal Standard Stock Number FSN 5330-196-6987), are intended for use on 2-inch round handholes of boiler economizers and superheaters installed on combatant naval ships and on economizers and waterwall headers installed on naval ship built for the U.S. Maritime Commission and converted to Navy use. These gaskets are applicable to Foster Wheeler Corporation boilers or economizers.

6.1.2 Size 1-1/4-inch gaskets (Federal Standard Stock Number FSN 5330-196-6988), are intended for use on 2-inch round handholes of boiler superheaters installed in ships built for the U.S. Maritime Commission and converted to Navy use. This size is applicable to superheaters of Foster Wheeler Corporation boilers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).
(d) Special marking, if required (see 5.3).

6.3 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Custodians:
Navy - SH
Air Force - 82

Preparing activity:
Navy - SH
(Project 5330-0284)

Review activities:
Navy - SH
Air Force - 82, 85

User activity:
Army - ME

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractors involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

(NAVSHIPS OVPRT 12-66)

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